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Smart Waste Classification using Edge AI

1. Aim of the Experiment

To develop a Smart Waste Classification system using Edge AI for real-time identification of wet and dry waste and provide visual indication using Arduino-controlled LEDs.

2. Introduction

Improper waste segregation at the source leads to environmental pollution and inefficient recycling. Manual classification methods are slow, inconsistent, and prone to errors, while cloud-based solutions suffer from latency and connectivity issues. Therefore, there is a need for a real-time, accurate, and low-cost system that can classify waste efficiently using Edge AI. Waste management is a critical issue due to increasing population and urbanization. Improper segregation of waste leads to environmental pollution and inefficient recycling. Manual classification methods are slow, inconsistent, and prone to errors.

With advancements in Artificial Intelligence and computer vision, automated waste classification systems have been developed. However, many existing systems rely on cloud computing, which introduces latency and requires continuous internet connectivity.

This project presents a **Smart Waste Classification system using Edge AI** that performs real-time classification of waste into wet and dry categories. The system uses image processing techniques and integrates with an Arduino Uno to provide LED-based visual indication. It offers a low-cost, efficient, and real-time solution for waste segregation.

3. Background / Literature Review

Waste management has become a critical global challenge due to rapid urbanization and increasing population. Improper segregation of waste at the source leads to environmental pollution, inefficient recycling, and health hazards. Traditional waste classification methods rely heavily on manual sorting, which is time-consuming, inconsistent, and prone to human error. These limitations highlight the need for automated and intelligent waste classification systems.

With the advancement of computer vision and Artificial Intelligence (AI), automated waste detection systems have gained significant attention. Earlier approaches used basic image processing techniques such as thresholding, color segmentation, and feature extraction. While

these methods are simple and cost-effective, they often lack robustness under varying lighting conditions and complex environments.

Recent developments in deep learning, particularly Convolutional Neural Networks (CNNs), have significantly improved object detection and classification accuracy. AI-based waste classification systems can automatically identify different categories of waste using trained models. However, most of these systems rely on cloud-based processing, which introduces challenges such as high latency, internet dependency, increased bandwidth usage, and potential privacy concerns.

To overcome these limitations, Edge AI has emerged as an effective solution. Edge AI enables data processing directly on local devices rather than relying on cloud servers. This approach reduces latency, enhances real-time performance, ensures data privacy, and minimizes network dependency. Edge-based systems are especially suitable for real-time applications such as waste classification, where immediate response is required.

Several existing edge-based systems focus primarily on software-level implementation without integrating hardware for physical output or action. Moreover, many solutions use complex and expensive hardware, making them less suitable for low-cost deployments.

The proposed system addresses these gaps by implementing a **real-time waste classification system using Edge AI** combined with **Arduino-based hardware integration**. The system utilizes image processing techniques through OpenCV to classify waste into wet and dry categories based on brightness analysis. The classification result is then communicated to an Arduino Uno, which provides visual feedback using LEDs. **Conclusion of Literature Survey (100 words)**

The literature review shows that waste classification has progressed from manual methods to AI-based automated systems. Traditional image processing techniques are simple but lack accuracy, while deep learning models improve performance but depend on cloud computing, leading to latency and connectivity issues. Edge AI emerges as an effective solution by enabling real-time, local processing with reduced delay and improved privacy. However, many existing systems lack hardware integration or are costly. The proposed system addresses these gaps by combining Edge AI with Arduino-based hardware, providing a low-cost, efficient, and real-time solution for basic waste classification with practical applicability and scope for future enhancement.

Aspect	Existing Manual Systems	Cloud-Based Systems	Proposed Edge AI System
Latency	High (human delay)	High (network delay)	Low
Internet Dependency	Not required	Required	Not required
Accuracy	Low	High	Moderate to High
Cost	Low	High	Low
Privacy	Secure	Risk of data exposure	Secure
Hardware Integration	Not available	Limited	Integrated (Arduino)
Real-Time Performance	Poor	Moderate	High

4. Objectives

- To develop a smart system for classifying waste into wet and dry categories
- To implement real-time waste detection using camera input and image processing
- To utilize Edge AI for fast and efficient local processing
- To interface software with Arduino Uno
- To provide visual indication using LEDs
- To design a low-cost solution

5 Components Required

5.1 Hardware Components

Table 1: Hardware Components Used

Component	Quantity	Purpose
Arduino Uno	1	Main controller
LED (Green)	1	Indicates wet waste
LED (Red)	1	Indicates dry waste
Resistor (220Ω)	2	Limits current for LEDs
USB Cable	1	Power and communication
Camera (via Python interface)	1	Captures real-time images

5.2 Software Components

Table 2: Software Components Used

Software	Purpose
Python	Programming and processing
OpenCV	Image processing and detection
NumPy	Numerical computations
PySerial	Communication with Arduino
Arduino IDE	Programming Arduino board

6 Methodology

The proposed system performs real-time waste classification using image processing and edge computing techniques. The complete workflow of the system is as follows:

- The system accesses the camera using a Python program to capture real-time frames.
- Motion detection is applied to identify the presence of an object in the frame.
- The captured image is processed using OpenCV by converting it into grayscale and analyzing its brightness.
- Based on the brightness threshold, the waste is classified as either wet or dry.
- The classification result is sent from the Python program to the Arduino Uno through serial communication.
- The Arduino processes the received signal and controls the LEDs accordingly:
 - Green LED indicates wet waste
 - Red LED indicates dry waste
- If no object is detected, both LEDs remain OFF.

This approach ensures real-time processing at the edge, resulting in low latency, faster response, and independence from internet connectivity.

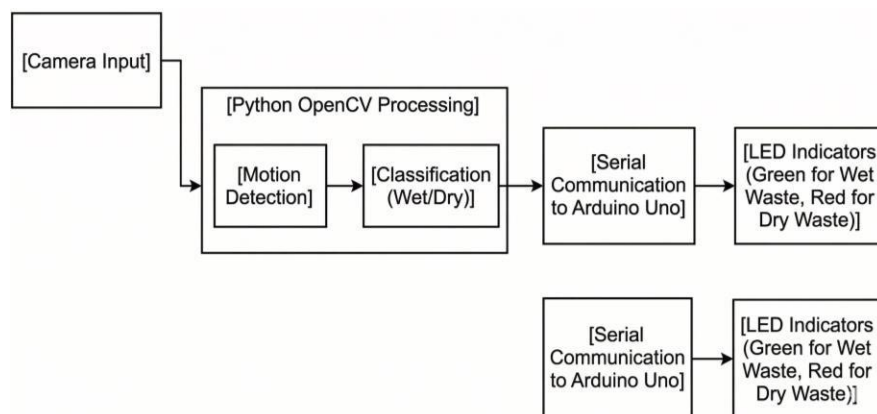


Figure 1: Block Diagram of Smart Waste Classification System

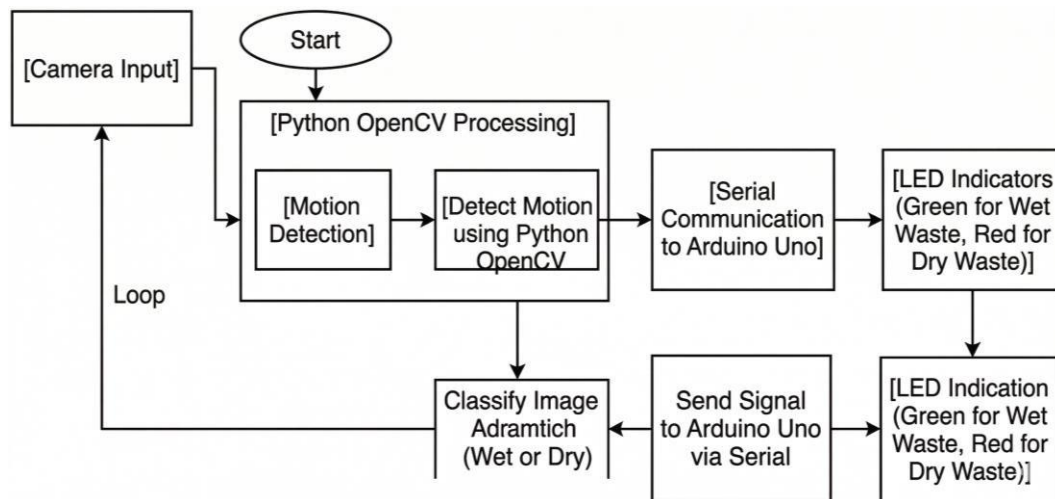


Figure 2: Flowchart of Waste Classification Process

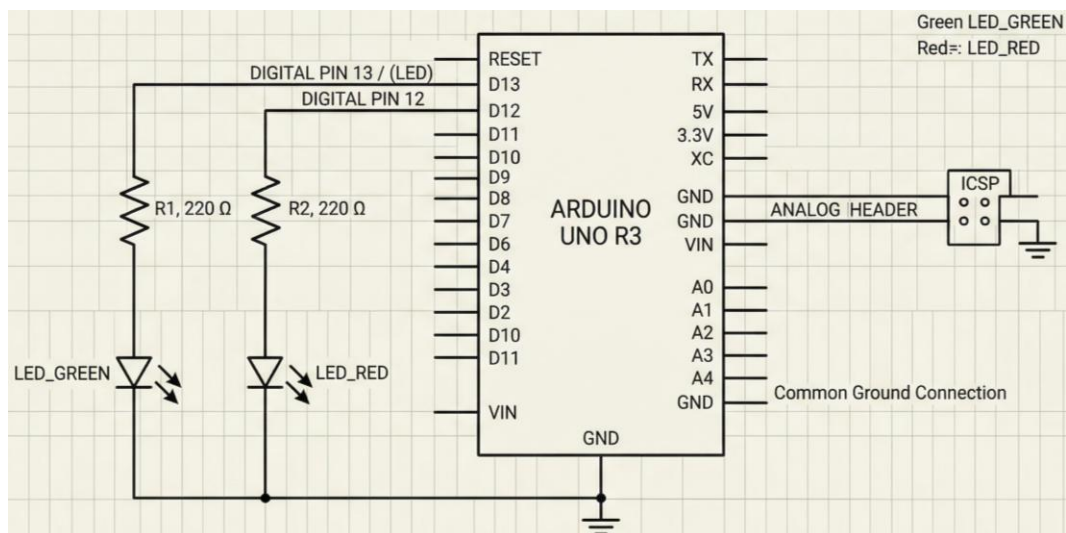


Figure 3: Circuit Diagram of Arduino with LED Indicators

7 Implementation

7.1 Camera Interface Code

```

import cv2
cap = cv2.VideoCapture(0)
while True:
    ret, frame = cap.read()
    cv2.imshow("Camera Test", frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
cap.release()
cv2.destroyAllWindows()
  
```

7.2 Waste Detection Code

```

import cv2
import serial
import time
import numpy as np
from collections import deque

arduino = serial.Serial('COM3', 9600)
time.sleep(2)
cap = cv2.VideoCapture(0)
prev_frame = None
prediction_buffer = deque(maxlen=10)
last_sent = None
last_detection_time = 0
HOLD_TIME = 2
def predict_waste(frame):
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    brightness = gray.mean()
    return "Wet" if brightness > 120 else "Dry"
while True:
    ret, frame = cap.read()
    if not ret:
        break
    frame = cv2.resize(frame, (224, 224))
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    gray_blur = cv2.GaussianBlur(gray, (21, 21), 0)
    if prev_frame is None:
        prev_frame = gray_blur
        continue
    frame_diff = cv2.absdiff(prev_frame, gray_blur)
    thresh = cv2.threshold(frame_diff, 25, 255, cv2.THRESH_BINARY)[1]
    motion = np.sum(thresh)
    current_time = time.time()
    if motion > 50000:
        prediction = predict_waste(frame)
        prediction_buffer.append(prediction)
        last_detection_time = current_time
    else:
        prediction_buffer.append("No")
        prev_frame = gray_blur

    if len(prediction_buffer) == 10:
        final_prediction = max(set(prediction_buffer), key=prediction_buffer.count)
        if current_time - last_detection_time < HOLD_TIME:
            if final_prediction == "Wet":
                waste = "Wet Waste"
                signal = b'W'
            elif final_prediction == "Dry":
                waste = "Dry Waste"
                signal = b'D'
            else:

```

```

        waste = "No Waste"
        signal = b'N'
    else:
        waste = "No Waste"
        signal = b'N'
        if signal != last_sent:
            arduino.write(signal)
            last_sent = signal
    else:
        waste = "Detecting..."
        display_frame = cv2.resize(frame, (640, 480))
        cv2.putText(display_frame, waste, (50, 50),
                    cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 255, 0), 2)
        cv2.imshow("Smart Waste Segregation", display_frame)
        if cv2.waitKey(1) & 0xFF == ord('q'):
            break
cap.release()
cv2.destroyAllWindows()
arduino.close()

```

7.3 Arduino Code

```

char data;
void setup() {
    pinMode(8, OUTPUT);
    pinMode(9, OUTPUT);
    Serial.begin(9600);
}
void loop() {
    if (Serial.available()) {
        data = Serial.read();
    }
    if (data == 'W') {
        digitalWrite(9, LOW);
        digitalWrite(8, HIGH);
        delay(300);
        digitalWrite(8, LOW);
        delay(300);
    }
    else if (data == 'D') {
        digitalWrite(8, LOW);
        digitalWrite(9, HIGH);
        delay(300);
        digitalWrite(9, LOW);
        delay(300);
    }
    else if (data == 'N') {
        digitalWrite(8, LOW);
        digitalWrite(9, LOW);
    }
}

```

}
}

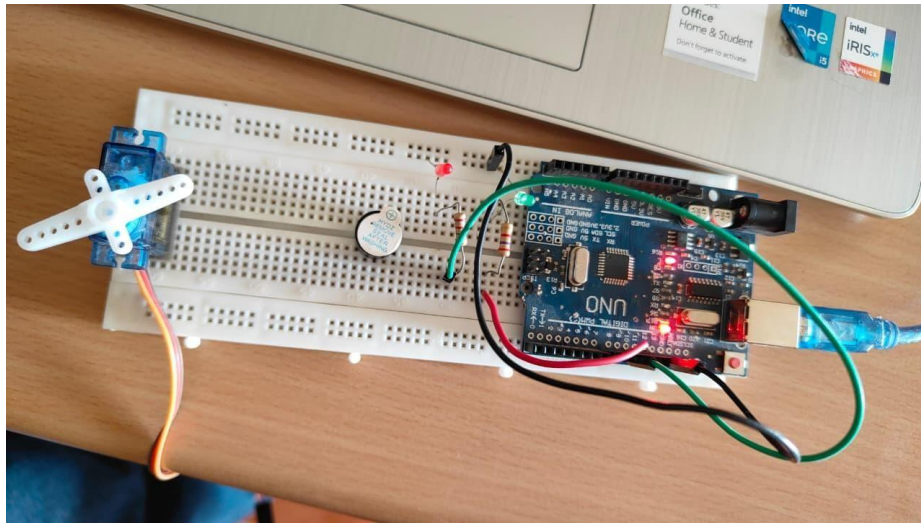


Figure 4: Hardware Setup of Smart Waste Classification System

8 Results and Analysis

8.1 Results

The developed system successfully performs real-time waste classification using image processing techniques. The classification results are indicated using LEDs connected to the Arduino Uno.

8.1.1 Green LED glows when wet waste is detected

8.1.2 Red LED glows when dry waste is detected

8.1.3 No LED glows when no object is present

8.1.4 The system responds in real-time with minimal delay

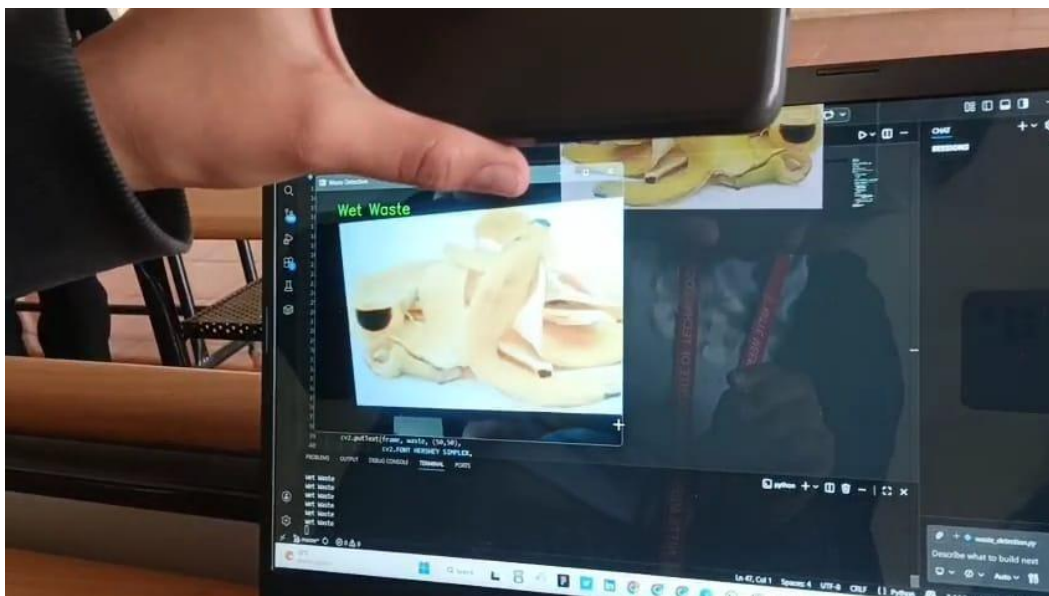


Figure 5: Detection of Wet Waste with Green LED Indication



Figure 6: Detection of Dry Waste with Red LED Indication

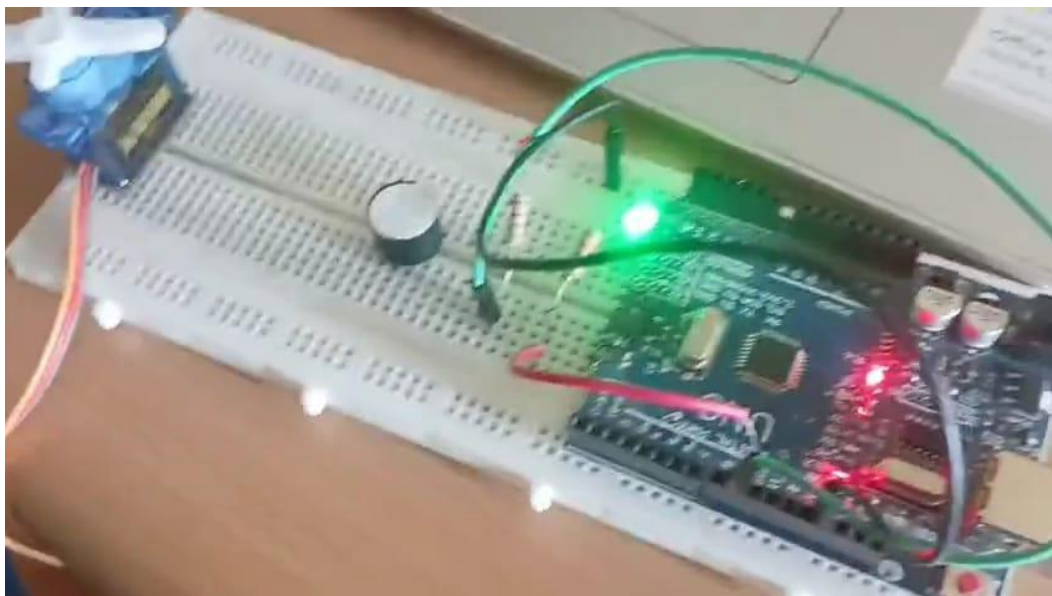


Figure 7: Hardware Output Showing LED Indication

8.2 Analysis

The system demonstrates the effectiveness of using Edge AI for real-time waste classification. The use of local processing reduces latency and eliminates dependency on internet connectivity. The motion detection mechanism ensures that classification occurs only when an object is present, improving efficiency.

However, the system relies on brightness-based classification, which may be affected by lighting conditions. Despite this limitation, the system provides a simple, low-cost, and practical solution for basic waste segregation.

9 Conclusion

The Smart Waste Classification system using Edge AI has been successfully developed and implemented. The system is capable of classifying waste into wet and dry categories in real-time using image processing techniques and controlling hardware through Arduino.

This project demonstrates a cost-effective and efficient approach for smart waste management. The use of Edge AI ensures faster processing, reduced latency, and independence from cloud-based systems.

10 Future Scope

- Implementation of deep learning models (CNN) for higher accuracy
- Integration with automatic waste segregation mechanisms
- Expansion to classify multiple waste categories
- Development of a compact embedded system for real-world deployment

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